AI4015 Agentic AI, Assignment Beta

Question 4: Evaluating an LLM Sentiment Classifier

Roll number: 23K-0008

Approach and settings

The LLM labels 300 customer reviews as Positive, Negative or Neutral. Stage 1 uses a zero-shot prompt; Stage 2 adds a definition of Neutral and six few-shot examples. All values below use the numbers given in the assignment, and every result was also produced by the spreadsheet and the notebook submitted with this report, so the three agree. Some choices change the answers, so I state them here:

- Confidence interval: Wilson score interval, z = 1.96 (95%).
- Standard deviation of the five runs: sample standard deviation with n − 1.
- BLEU-4: uniform weights (0.25 each), tokens split on spaces, no smoothing. For this pair no n-gram precision is zero, so smoothing would not change the score (NLTK method 1 gives the identical value).
- ROUGE: ROUGE-1, ROUGE-2 and ROUGE-L F1, no stemming, computed by hand and checked with the rouge-score package.
- METEOR: NLTK implementation with alpha = 0.9, beta = 3, gamma = 0.5, exact and Porter-stem matching. The WordNet synonym stage is switched off, because the WordNet data could not be downloaded in my environment, so I cannot say whether synonym matching would change the result.
- Logarithms are natural logarithms; values are kept to six decimal places.

1. Stage 1: overall accuracy and 95% confidence interval

Table. Stage 1 confusion matrix (rows = actual, columns = predicted).

Actual / Predicted	**Positive**	**Negative**	**Neutral**	**Row total**
Positive	100	5	15	120
Negative	8	80	12	100
Neutral	20	10	50	80
Column total	128	95	77	300

Accuracy = (number of correct predictions) / (total samples) = trace of the matrix / n.

$$\text{Correct} = 100 + 80 + 50 = 230, \quad n = 300$$

$$\text{Accuracy} = 230 / 300 = 0.766667 = 76.67\%$$

95% confidence interval, Wilson score method:

$$\text{lower, upper} = [\ \hat{p} + z^2/(2n) \pm z \cdot \sqrt{(\ \hat{p}(1 - \hat{p})/n + z^2/(4n^2)\ )}\ ] / (\ 1 + z^2/n\ )$$

$$n = 300, \quad \hat{p} = 0.766667, \quad z = 1.96, \quad z^2 = 3.841600$$

$$z^2/n = 3.841600 / 300 = 0.012805; \quad 1 + z^2/n = 1.012805$$

$$\hat{p} + z^2/(2n) = 0.766667 + 0.006403 = 0.773069; \quad \text{centre} = 0.773069 / 1.012805 = 0.763295$$

$$\hat{p}(1 - \hat{p})/n = 0.766667 \times 0.233333 / 300 = 0.00059630; \quad z^2/(4n^2) = 3.841600 / 360000 = 0.00001067$$

$$\sqrt{(0.00059630 + 0.00001067)} = \sqrt{0.00060697} = 0.024637; \quad z \times \text{that} = 0.048288$$

$$\text{half-width} = 0.048288 / 1.012805 = 0.047677$$

$$\text{Lower} = 0.763295 - 0.047677 = 0.715618; \quad \text{Upper} = 0.763295 + 0.047677 = 0.810973$$

95% CI for accuracy: [0.7156, 0.8110], that is 71.56% to 81.10%.

2. Stage 1: precision, recall and F1 for each class

Precision = TP/(TP + FP), Recall = TP/(TP + FN), F1 = 2PR/(P + R).

Positive: TP = 100 (diagonal); FP = column total − TP = 128 − 100 = 28; FN = row total − TP = 120 − 100 = 20.

$$\text{Precision} = 100 / (100 + 28) = 100 / 128 = 0.781250$$

$$\text{Recall} = 100 / (100 + 20) = 100 / 120 = 0.833333$$

$$F1 = 2 \times 0.781250 \times 0.833333 / (0.781250 + 0.833333) = 1.302083 / 1.614583 = 0.806452$$

Negative: TP = 80 (diagonal); FP = column total − TP = 95 − 80 = 15; FN = row total − TP = 100 − 80 = 20.

$$\text{Precision} = 80 / (80 + 15) = 80 / 95 = 0.842105$$

$$\text{Recall} = 80 / (80 + 20) = 80 / 100 = 0.800000$$

$$F1 = 2 \times 0.842105 \times 0.800000 / (0.842105 + 0.800000) = 1.347368 / 1.642105 = 0.820513$$

Neutral: TP = 50 (diagonal); FP = column total − TP = 77 − 50 = 27; FN = row total − TP = 80 − 50 = 30.

$$\text{Precision} = 50 / (50 + 27) = 50 / 77 = 0.649351$$

$$\text{Recall} = 50 / (50 + 30) = 50 / 80 = 0.625000$$

$$F1 = 2 \times 0.649351 \times 0.625000 / (0.649351 + 0.625000) = 0.811688 / 1.274351 = 0.636943$$

Table. Stage 1 per-class results.

Class	TP	FP	FN	Support	Precision	Recall	F1
Positive	100	28	20	120	0.781250	0.833333	0.806452
Negative	80	15	20	100	0.842105	0.800000	0.820513
Neutral	50	27	30	80	0.649351	0.625000	0.636943

3. Stage 1: macro, weighted and micro F1

Macro F1 = (F1_pos + F1_neg + F1_neu) / 3

Macro F1 = (0.806452 + 0.820513 + 0.636943) / 3 = 2.263907 / 3 = 0.754636

Weighted F1 = Σ (support_k × F1_k) / n

= (120 × 0.806452 + 100 × 0.820513 + 80 × 0.636943) / 300

= (96.774194 + 82.051282 + 50.955414) / 300 = 229.780890 / 300 = 0.765936

Micro F1 uses the totals over all classes:

ΣTP = 230, ΣFP = 70, ΣFN = 70

Micro precision = 230 / (230 + 70) = 0.766667; Micro recall = 230 / (230 + 70) = 0.766667

Micro F1 = 2 × 0.766667 × 0.766667 / (0.766667 + 0.766667) = 0.766667

Micro F1 equals accuracy (0.766667) here. Each review has exactly one true label and one predicted label, so every mistake is counted once as a false positive (for the predicted class) and once as a false negative (for the true class). That makes ΣFP = ΣFN = 70, so micro precision = micro recall = ΣTP/n, which is accuracy.

Table. Stage 1 averaged F1 scores.

Average	**Value**
Macro F1	0.754636
Weighted F1	0.765936
Micro F1	0.766667

4. Interpretation of Stage 1

Overall the zero-shot model is right on 230 of 300 reviews (76.67%), with a wide interval of about 71.6% to 81.1% because n is only 300. By F1, Negative is classified best (0.8205, with the highest precision, 0.8421: of the 95 reviews predicted Negative, 80 really are). Positive has the highest recall (100 of 120, 0.8333) and a close F1 of 0.8065. Neutral is clearly the weakest, with precision 0.6494, recall 0.6250 and F1 0.6369.

The matrix shows where the mistakes go. Neutral reviews are sent to Positive 20 times (25% of all Neutral reviews) and to Negative 10 times, so only 50 of 80 survive. The Positive class loses 15 of its 120 reviews to Neutral and the Negative class loses 12 of 100. Mistakes between the two opposite classes are small by comparison: 5 Positive reviews predicted as Negative and 8 Negative predicted as Positive. So the model separates good from bad reasonably well and fails mainly at the boundary with Neutral. The Positive predictions are also polluted by Neutral: 20 of the 28 false Positives are really Neutral reviews.

Neutral is hard for a prompt that only supplies the label names. The model has to guess what the word means: a factual review, a mixed one ("late delivery but good quality") and a review with no opinion are different things, and a model that sees a positive phrase tends to commit to Positive. The 27 Positive and Negative reviews pushed into Neutral (15 + 12) suggest it also falls back on Neutral when the review is mixed. Recommended changes: (1) define Neutral explicitly as factual, mixed or no clear opinion; (2) add a few labelled examples per

class, including borderline mixed reviews; (3) ask for a one-sentence justification so that wrong labels can be inspected; (4) test different example orders for stability; (5) keep a small set of Neutral errors for later error analysis. Stage 2 applies the first two.

5. Stage 2 metrics and comparison with Stage 1

Table. Stage 2 confusion matrix (rows = actual, columns = predicted).

Actual / Predicted	**Positive**	**Negative**	**Neutral**	**Row total**
Positive	106	4	10	120
Negative	6	84	10	100
Neutral	9	7	64	80
Column total	121	95	84	300

Accuracy = (number of correct predictions) / (total samples) = trace of the matrix / n.

$$\text{Correct} = 106 + 84 + 64 = 254, \quad n = 300$$

$$\text{Accuracy} = 254 / 300 = 0.846667 = 84.67\%$$

95% confidence interval, Wilson score method:

$$\text{lower, upper} = [\ \hat{p} + z^2/(2n) \pm z \cdot \sqrt{(\ \hat{p}(1 - \hat{p})/n + z^2/(4n^2)\ )}\ ] / (\ 1 + z^2/n\ )$$

$$n = 300, \quad \hat{p} = 0.846667, \quad z = 1.96, \quad z^2 = 3.841600$$

$$z^2/n = 3.841600 / 300 = 0.012805; \quad 1 + z^2/n = 1.012805$$

$$\hat{p} + z^2/(2n) = 0.846667 + 0.006403 = 0.853069; \quad \text{centre} = 0.853069 / 1.012805 = 0.842284$$

$$\hat{p}(1 - \hat{p})/n = 0.846667 \times 0.153333 / 300 = 0.00043274; \quad z^2/(4n^2) = 3.841600 / 360000 = 0.00001067$$

$$\sqrt{(0.00043274 + 0.00001067)} = \sqrt{0.00044341} = 0.021057; \quad z \times \text{that} = 0.041272$$

$$\text{half-width} = 0.041272 / 1.012805 = 0.040751$$

$$\text{Lower} = 0.842284 - 0.040751 = 0.801533; \quad \text{Upper} = 0.842284 + 0.040751 = 0.883034$$

95% CI for accuracy: [0.8015, 0.8830], that is 80.15% to 88.30%.

Stage 2 per-class results:

Precision = TP/(TP + FP), Recall = TP/(TP + FN), F1 = 2PR/(P + R).

Positive: TP = 106 (diagonal); FP = column total − TP = 121 − 106 = 15; FN = row total − TP = 120 − 106 = 14.

$$\text{Precision} = 106 / (106 + 15) = 106 / 121 = 0.876033$$

$$\text{Recall} = 106 / (106 + 14) = 106 / 120 = 0.883333$$

$$F1 = 2 \times 0.876033 \times 0.883333 / (0.876033 + 0.883333) = 1.547658 / 1.759366 = 0.879668$$

Negative: TP = 84 (diagonal); FP = column total − TP = 95 − 84 = 11; FN = row total − TP = 100 − 84 = 16.

$$\text{Precision} = 84 / (84 + 11) = 84 / 95 = 0.884211$$

$$\text{Recall} = 84 / (84 + 16) = 84 / 100 = 0.840000$$

$$F1 = 2 \times 0.884211 \times 0.840000 / (0.884211 + 0.840000) = 1.485474 / 1.724211 = 0.861538$$

Neutral: TP = 64 (diagonal); FP = column total − TP = 84 − 64 = 20; FN = row total − TP = 80 − 64 = 16.

$$\text{Precision} = 64 / (64 + 20) = 64 / 84 = 0.761905$$

$$\text{Recall} = 64 / (64 + 16) = 64 / 80 = 0.800000$$

$$F1 = 2 \times 0.761905 \times 0.800000 / (0.761905 + 0.800000) = 1.219048 / 1.561905 = 0.780488$$

Table. Stage 2 per-class results.

Class	**TP**	**FP**	**FN**	**Support**	**Precision**	**Recall**	**F1**
Positive	106	15	14	120	0.876033	0.883333	0.879668
Negative	84	11	16	100	0.884211	0.840000	0.861538
Neutral	64	20	16	80	0.761905	0.800000	0.780488

Macro F1 = (F1_pos + F1_neg + F1_neu) / 3

$$\text{Macro F1} = (0.879668 + 0.861538 + 0.780488) / 3 = 2.521694 / 3 = 0.840565$$

Weighted F1 = Σ (support_k × F1_k) / n

$$= (120 \times 0.879668 + 100 \times 0.861538 + 80 \times 0.780488) / 300$$

$$= (105.560166 + 86.153846 + 62.439024) / 300 = 254.153037 / 300 = 0.847177$$

Micro F1 uses the totals over all classes:

$$\Sigma TP = 254, \quad \Sigma FP = 46, \quad \Sigma FN = 46$$

Micro precision = 254 / (254 + 46) = 0.846667; Micro recall = 254 / (254 + 46) = 0.846667

$$\text{Micro F1} = 2 \times 0.846667 \times 0.846667 / (0.846667 + 0.846667) = 0.846667$$

Micro F1 equals accuracy (0.846667) here. Each review has exactly one true label and one predicted label, so every mistake is counted once as a false positive (for the predicted class) and once as a false negative (for the true class). That makes ΣFP = ΣFN = 46, so micro precision = micro recall = ΣTP/n, which is accuracy.

Table. Stage 2 averaged F1 scores.

Average	**Value**
Macro F1	0.840565
Weighted F1	0.847177
Micro F1	0.846667

Table. Stage 1 versus Stage 2.

Metric	**Stage 1**	**Stage 2**	**Change**
Accuracy	0.766667	0.846667	+0.080000
Positive Precision	0.781250	0.876033	+0.094783
Positive Recall	0.833333	0.883333	+0.050000
Positive F1	0.806452	0.879668	+0.073216
Negative Precision	0.842105	0.884211	+0.042105
Negative Recall	0.800000	0.840000	+0.040000
Negative F1	0.820513	0.861538	+0.041026
Neutral Precision	0.649351	0.761905	+0.112554
Neutral Recall	0.625000	0.800000	+0.175000
Neutral F1	0.636943	0.780488	+0.143545
Macro F1	0.754636	0.840565	+0.085929
Weighted F1	0.765936	0.847177	+0.081240
Micro F1	0.766667	0.846667	+0.080000

Every metric in the table went up, so for these data the revised prompt is better on every measure I computed. Accuracy rose from 0.7667 to 0.8467 (+0.0800, 24 more correct reviews: 254 against 230). The biggest gain is for Neutral, which was the weakest class: recall +0.1750 (0.6250 to 0.8000), precision +0.1126 and F1 +0.1435 (0.6369 to 0.7805). That matches the intended effect of defining Neutral and showing borderline examples; Neutral reviews wrongly sent to Positive dropped from 20 to 9. The smallest gains are for the Negative class (precision +0.0421, recall +0.0400, F1 +0.0410) and for Positive recall (+0.0500). Macro F1 gained the most of the three averages (+0.0859) because it weights Neutral as much as the other classes; weighted and micro F1 gained +0.0812 and +0.0800.

A caution on the size of the improvement: the two 95% intervals ([0.7156, 0.8110] and [0.8015, 0.8830]) overlap slightly, and both stages were run on the same 300 reviews. Overlapping intervals do not prove the difference is not real, and a proper significance test on paired data (for example McNemar's test) would need the review-level predictions, which were not given. So I report the gain as large and consistent across all metrics, but I do not claim statistical significance.

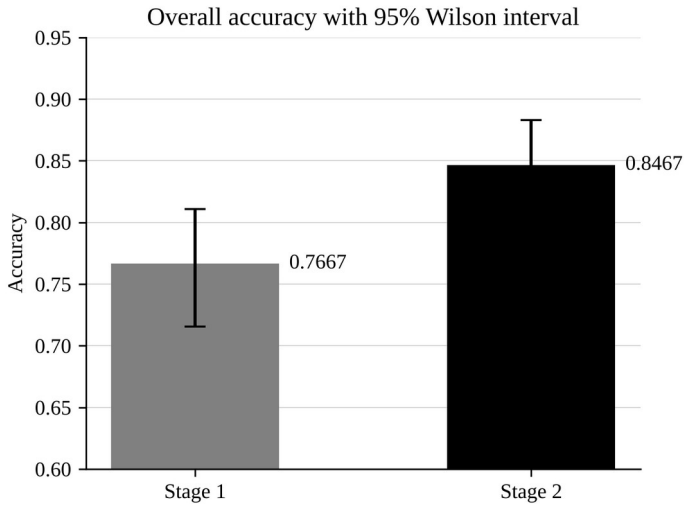


Figure 1. Overall accuracy with 95% Wilson intervals.

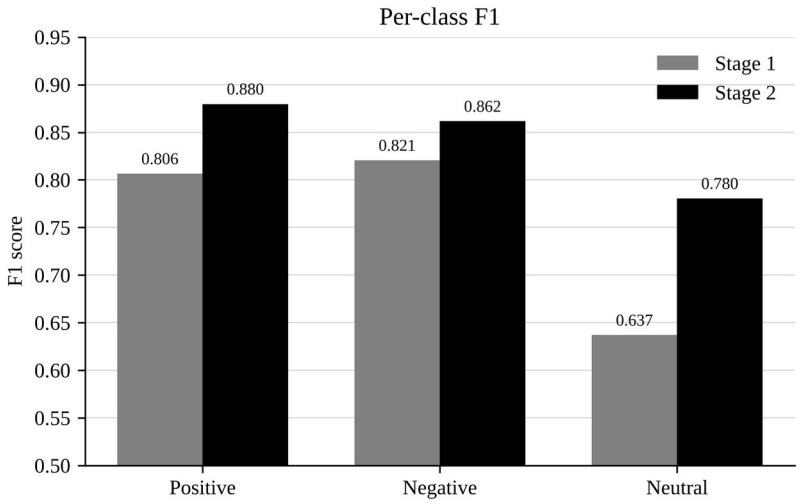


Figure 2. Per-class F1, Stage 1 and Stage 2.

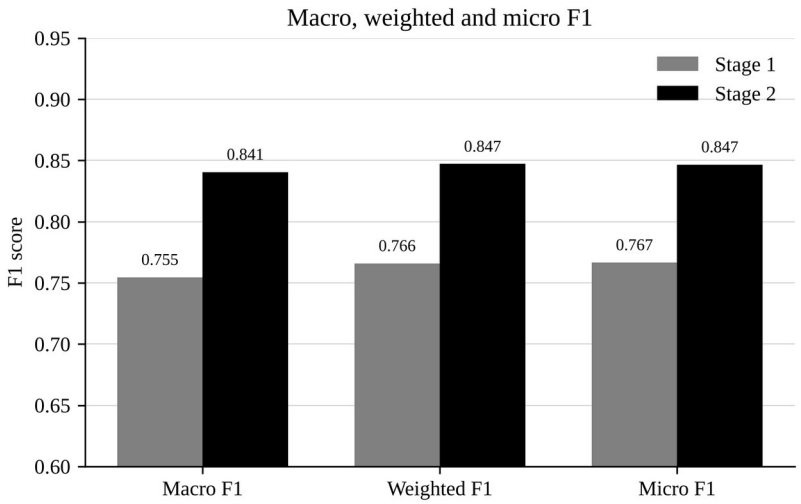


Figure 3. Macro, weighted and micro F1.

6. Few-shot prompt stability

Five runs with different example orderings gave accuracies 0.847, 0.840, 0.853, 0.837, 0.850. I use the sample standard deviation (divide by n − 1 = 4), because these five runs are a sample of the possible orderings.

$$\text{mean} = \Sigma x / n, \quad s = \sqrt{\Sigma(x - \text{mean})^2 / (n - 1)}$$

$$\Sigma x = 0.847 + 0.840 + 0.853 + 0.837 + 0.850 = 4.227$$

$$\text{mean} = 4.227 / 5 = 0.8454$$

Table. Deviations from the mean.

Run	**x**	**x − mean**	**(x − mean)²**
1	0.847	+0.0016	0.00000256
2	0.840	-0.0054	0.00002916
3	0.853	+0.0076	0.00005776
4	0.837	-0.0084	0.00007056
5	0.850	+0.0046	0.00002116
Sum	4.227	0.0000	0.00018120

$$\Sigma(x - \text{mean})^2 = 0.00018120$$

$$\text{variance} = 0.00018120 / 4 = 0.00004530$$

$$s = \sqrt{0.00004530} = 0.006731$$

Result: mean ± standard deviation = 0.8454 ± 0.0067 (84.54% ± 0.67 percentage points).

The five runs lie between 0.837 and 0.853, a spread of only 1.6 percentage points, and the standard deviation (0.67 points) is small next to the 4-point half-width of the confidence interval. The prompt is therefore reasonably stable across example orderings. The mean (0.8454) is also very close to the single Stage 2 accuracy of 0.8467, so that figure was not a lucky ordering. Five runs is a small number, so this is a rough estimate of the variation.

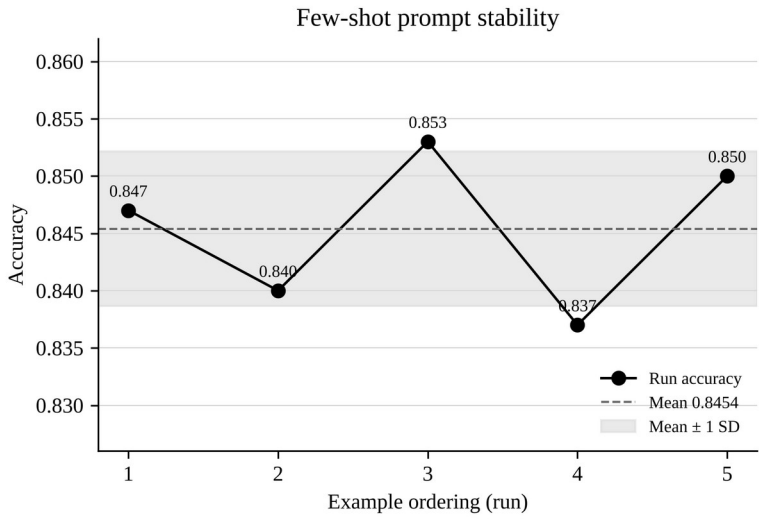


Figure 4. Accuracy across five example orderings (shaded band: mean ± 1 SD).

7. BLEU-4, ROUGE-1, ROUGE-2, ROUGE-L and METEOR

Reference (10 tokens): the delivery was late but the product quality is good

Candidate (9 tokens): the product quality is good but delivery was slow

Both strings are already lower-cased without punctuation, so tokens are obtained by splitting on spaces.

BLEU-4 measures modified (clipped) n-gram precision of the candidate against the reference for n = 1 to 4, combined by a geometric mean and multiplied by a brevity penalty. It is precision oriented.

Clipped n-gram matches (a candidate n-gram counts at most as often as it appears in the reference):

1-grams: the, product, quality, is, good, but, delivery, was match; 'slow' does not → 8 / 9

2-grams: the product, product quality, quality is, is good, delivery was match → 5 / 8

3-grams: the product quality, product quality is, quality is good match → 3 / 7

4-grams: the product quality is, product quality is good match → 2 / 6

$p_1 = 0.888889$, $p_2 = 0.625000$, $p_3 = 0.428571$, $p_4 = 0.333333$

ln p = -0.117783, -0.470004, -0.847298, -1.098612; sum = -2.533697; mean = -0.633424; geometric mean = exp(-0.633424) = 0.530771

Brevity penalty: candidate length c = 9 < reference length r = 10, so BP = exp(1 − r/c) = exp(1 − 10/9) = 0.894839

BLEU-4 = BP × geometric mean = 0.894839 × 0.530771 = 0.474955

Smoothing: none was needed, because every precision is above zero (the usual problem with short sentences does not arise here); NLTK's method 1 smoothing gives the same value. BLEU-4 = 0.4750, which is a moderate score: the sentences share many 1- to 4-grams, but the two clauses are in reversed order, which breaks the longer matches.

ROUGE-N measures n-gram overlap, mainly recall of the reference; ROUGE-L uses the longest common subsequence (LCS). I report precision, recall and F1 (the harmonic mean).

ROUGE-1: overlap 8; P = 8/9 = 0.888889, R = 8/10 = 0.800000, F1 = 0.842105

ROUGE-2: overlap 5 (reference has 9 bigrams, candidate 8); P = 5/8 = 0.625000, R = 5/9 = 0.555556, F1 = 0.588235

ROUGE-L: LCS = 'the product quality is good' = 5 words; P = 5/9 = 0.555556, R = 5/10 = 0.500000, F1 = 0.526316

ROUGE-1 is high because eight of the nine candidate words appear in the reference. ROUGE-2 and ROUGE-L fall to about 0.59 and 0.53 because the clause order is swapped: the reference starts with the delivery and ends with the quality, the candidate does the opposite, so the longest in-order match is only the five-word quality phrase.

METEOR aligns candidate and reference words (exact match, then stem match, then synonyms if enabled), computes a recall-weighted F-mean and subtracts a fragmentation penalty for how many separate chunks the matches form. Settings: NLTK, alpha = 0.9, beta = 3, gamma = 0.5, exact and Porter-stem matching, WordNet disabled.

matched words m = 8: the, product, quality, is, good, but, delivery, was ('slow' and 'late' are left unmatched; the stem stage adds no matches)

P = 8/9 = 0.888889, R = 8/10 = 0.800000

Fmean = P·R / (0.9·P + 0.1·R) = 0.711111 / 0.880000 = 0.808081

Chunks: (the product quality is good) + (but) + (delivery was) = 3 chunks

Penalty = 0.5 × (chunks/m)³ = 0.5 × (3/8)³ = 0.5 × 0.052734 = 0.026367

METEOR = Fmean × (1 − penalty) = 0.808081 × 0.973633 = 0.786774

METEOR = 0.7868. The NLTK function returns the same value (0.786774), which I checked in the notebook. It scores higher than BLEU because it is tolerant of reordering: only three chunks are needed to cover the matches.

Table. Overlap metrics for the generated justification.

Metric	**Score**
BLEU-4 (no smoothing)	0.474955
ROUGE-1 F1	0.842105
ROUGE-2 F1	0.588235
ROUGE-L F1	0.526316
METEOR (no WordNet)	0.786774

Why these scores miss the meaning: the candidate says almost the same thing as the reference (late and slow are near-synonyms, and the two clauses are only reordered), yet BLEU and ROUGE-2 are heavily lowered by the word order and none of the metrics gives credit for 'slow' against 'late'. They compare surface strings, not meaning. METEOR could give credit for synonyms through WordNet, but I did not run that stage, so I cannot claim it would link these two words. A single sentence pair is also too little to draw general conclusions about the model.

8. Perplexity of the generated justification

Perplexity = exp(−(1/N) Σ ln p_i), where p_i are the probabilities the model gave to its N = 9 generated tokens.

Table. Token probabilities and their natural logarithms.

i	**Token**	**p_i**	**ln p_i**
1	the	0.60	-0.510826
2	product	0.40	-0.916291
3	quality	0.50	-0.693147
4	is	0.80	-0.223144
5	good	0.70	-0.356675
6	but	0.30	-1.203973
7	delivery	0.25	-1.386294
8	was	0.50	-0.693147
9	slow	0.20	-1.609438

$$\Sigma \ln p_i = (-0.510826) + (-0.916291) + (-0.693147) + (-0.223144) + (-0.356675) + (-1.203973) + (-1.386294) + (-0.693147) + (-1.609438)$$

$$\Sigma \ln p_i = -7.592934$$

$$-(1/N)\,\Sigma \ln p_i = 7.592934 / 9 = 0.843659$$

$$\text{Perplexity} = \exp(0.843659) = 2.324859$$

Perplexity ≈ 2.3249. On average the model was about as uncertain as if it were choosing between roughly 2.3 equally likely tokens at each step; lower is better and 1 would mean complete certainty. The least probable tokens were 'slow' (0.20) and 'delivery' (0.25). This number describes how confident the language model was about this one sentence. It says nothing about classification accuracy, which is measured separately by the confusion matrices.